

Toeplitz Covariance Estimation Under Coarse Quantization and Sparse Sampling

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Abstract—This paper addresses the estimation of a Hermitian Toeplitz covariance matrix under sparse observations and coarse quantization. Within the triangular quantization framework, we propose an unbiased estimator that accounts for quantization effects, termed the quantized Toeplitz-projected sample covariance matrix (Q-TSCM). Further, based on a covariance-fitting criterion, we propose the quantized sparse and parametric approach (Q-SPA), which additionally enforces positive semidefiniteness. Numerical experiments are presented to demonstrate the effectiveness of the proposed estimators.

Index Terms—Covariance matrix estimation, Toeplitz covariance, multi-bit quantization, dithering

I. INTRODUCTION

In multivariate statistics [1] and signal processing [2], accurate covariance estimation directly impacts the performance of subsequent detection, estimation, and inference tasks. In particular, in direction-of-arrival (DOA) estimation [3] and radar image processing [4], it is commonly assumed that the data satisfies (generalized) stationarity, meaning that the covariance between two measurements depends solely on their separation. This, in turn, motivates modeling the covariance matrix $\mathbf{T} \in \mathbb{C}^{d \times d}$ as a Hermitian Toeplitz matrix.

However, practical sensing systems are often subject to two fundamental constraints: *incomplete spatial sampling* and *limited quantization precision*. On the one hand, in practical applications such as DOA estimation, fully sampled spatial observations are often obtained using uniform linear arrays (ULAs). Deploying large-scale ULAs may incur not only high hardware costs but also increased susceptibility to mutual coupling effects [5], [6]. As a result, sparse linear arrays (SLAs) are often employed in engineering practice [7], implying that only a sparse subset of array positions is observed.

On the other hand, practical systems must account for the effects of analogue-to-digital conversion (ADC), i.e., the quantization of continuous-valued observations. High-resolution ADCs are typically costly and power-hungry [8]. Consequently, in large-scale or low-power systems, subsequent processing and inference tasks are often carried out using coarsely quantized data [9], [10], which can also offer significant advantages in terms of data storage, transmission, and processing costs. Thus, another central concern is how to reduce signal processing costs via coarse quantization

while preserving covariance estimation accuracy as much as possible.

This paper addresses the problem of estimating Hermitian Toeplitz covariance matrices under practical hardware constraints of coarse quantization and sparse sampling. Within the triangular quantization framework, we propose an unbiased estimator, termed the quantized Toeplitz-projected sample covariance matrix (Q-TSCM). Moreover, based on covariance-fitting criteria [11], [12], we propose a quantized sparse and parametric approach (Q-SPA) that additionally enforces positive semidefiniteness, at the cost of solving a semidefinite program (SDP).

Research on Toeplitz covariance estimation from infinite-precision measurements is well established. In practice, incorporating a Toeplitz prior can facilitate covariance estimation under sparse observations [13], [14]. For instance, the ruler-based method recovers the full covariance matrix by applying a Toeplitz projection to the sub-sample covariance matrix, thereby accounting for sparse sampling [1], [14]. Furthermore, sparse recovery schemes based on the Vandermonde decomposition of $\mathbf{T}$ [15], sparse Fourier transforms [1], covariance-fitting criteria [11], [12], and other algorithmic approaches [4], [16]–[18] have been widely studied. However, these methods do not explicitly account for the additional bias induced by coarse quantization, making them non-trivial to apply in our setting.

In contrast, existing quantized covariance estimation schemes primarily focus on how quantization affects the sample covariance matrix, e.g., [13], [19]–[24]. Most of these works do not explicitly incorporate additional structural priors, which limits their applicability in sparse-observation settings. The prior work [14] studied real-valued Toeplitz covariance estimation from incomplete quantized data within the triangular-dithering quantization framework introduced in [23]. However, extending these results to the complex domain is non-trivial. In particular, a naive real-lifting approach yields a covariance matrix that is only block Toeplitz, making the Toeplitz-prior-based theory developed in [14] not directly applicable.

The remainder of this paper is organized as follows. Section II briefly introduces triangular quantization, based on which we present the proposed method in Section III and validate its effectiveness through numerical experiments in Section IV.

Notation: For a positive integer d we write $[d] = \{0, \dots, d-1\}$. For vectors and matrices, we use $\|\cdot\|_2$ to denote the

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Euclidean norm (for vectors) and the operator norm (for matrices), respectively. We write $A \succeq B$ if $A - B$ is positive semidefinite. We denote by $\mathcal{T}(\gamma)$ the Hermitian Toeplitz covariance matrix generated by the sequence γ . $Q_\Delta(\cdot)$ denotes the element-wise quantization operator with level Δ . Symbols with a hat (e.g., $\hat{T}$) denote covariance estimators constructed from quantized data, while symbols with a dot (e.g., $\dot{z}$) denote the corresponding quantized observations.

II. TRIANGULAR QUANTIZATION SCHEME

For a complex signal $z \in \mathbb{C}^d$, the *triangular quantization* is defined by

$$\dot{z} = Q_{\Delta, \mathbb{C}}(z + \tau), \quad (1)$$

where $\tau \in \mathbb{C}^d$ is a random dither vector, independent of z , whose entries are i.i.d. samples drawn from triangular distribution, i.e., $\tau_j = \Re(\tau_j) + i \cdot \Im(\tau_j)$ and

$$\Re(\tau_j), \Im(\tau_j) \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}\left(\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]\right) + \mathcal{U}\left(\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]\right) \quad (2)$$

for $j = 1, 2, \dots, d$. The operator $Q_{\Delta, \mathbb{C}}(\cdot)$ denotes entry-wise quantization of complex-valued data and is given by

$$Q_{\Delta, \mathbb{C}}(z) := Q_{\Delta, \mathbb{R}}(\Re(z)) + i \times Q_{\Delta, \mathbb{R}}(\Im(z)). \quad (3)$$

For a real number x , the memoryless scalar quantizer $Q_{\Delta, \mathbb{R}}(\cdot)$ is defined by

$$\dot{x} = Q_{\Delta, \mathbb{R}}(x) = \Delta \left(\left\lfloor \frac{x}{\Delta} \right\rfloor + \frac{1}{2} \right). \quad (4)$$

To ensure that (4) is well defined for $\Delta = 0$, we additionally set $Q_{0, \mathbb{R}}(x) = x$. The quantization scheme (1) generalizes the real-valued setting considered in [14], [22]–[24] to the complex domain. We refer to Δ as the *quantization level*, where a larger Δ corresponds to a lower quantization resolution.

Applying a triangular dither prior to quantization is crucial. Without dithering, distinct original covariance matrices can collapse to the same second-order statistics of the quantized data, thereby hindering recovery of the true covariance matrix from quantized measurements. On the other hand, incorporating a triangular dither can endow the quantized measurements with desirable statistical properties for covariance estimation. To illustrate this, we consider a real-valued signal $x \in \mathbb{R}^d$. For a prescribed quantization level Δ , under triangular quantization, it can be shown that the second-order moment of the quantization noise $\xi := \dot{x} - x$ does not depend on x ¹, and it satisfies [14], [23], [25]

$$\mathbb{E}[\dot{x}\dot{x}^T] = \mathbb{E}[xx^T] + \frac{\Delta^2}{4}\mathbf{I}_d. \quad (5)$$

This makes triangular quantization a natural choice for quantized covariance estimation.

¹In general, ξ and x are not independent.

III. THE PROPOSED ALGORITHM

We assume that $z \sim \mathcal{D}$ is a d -dimensional complex-valued random vector with covariance matrix T , and that T is *positive semidefinite Hermitian Toeplitz*, i.e.,

$$T = \mathcal{T}(\gamma) = [\gamma_{j-k}]_{1 \leq j, k \leq d},$$

where $\gamma_{-s} = \gamma_s^*$ to ensure Hermitian symmetry. Let $z^{(1)}, z^{(2)}, \dots, z^{(n)}$ be i.i.d. copies of z . For a prescribed quantization level $\Delta > 0$ and a given set of (sparsely) located observation indices Ω , we observe the quantized samples $\dot{z}_\Omega^{(l)}$, $l = 1, 2, \dots, n$. We are interested in estimating the true covariance matrix T from the partially observed, coarsely quantized data $\{\dot{z}_\Omega^{(l)}\}_{l=1}^n$.

A. Q-TSCM

We write $\dot{z}^{(l)} = \dot{x}^{(l)} + i\dot{y}^{(l)}$ and $z^{(l)} = x^{(l)} + iy^{(l)}$. Since the real and imaginary parts are quantized independently, it follows from (5) that

$$\begin{aligned} & \mathbb{E}[\dot{z}^{(l)} \dot{z}^{(l)H}] \\ &= \mathbb{E}[\dot{x}^{(l)} \dot{x}^{(l)H}] + \mathbb{E}[\dot{y}^{(l)} \dot{y}^{(l)H}] \\ & \quad + i \left(\mathbb{E}[\dot{x}^{(l)} \dot{y}^{(l)H}] + \mathbb{E}[\dot{y}^{(l)} \dot{x}^{(l)H}] \right) \\ &= \mathbb{E}[x^{(l)} x^{(l)H}] + \frac{\Delta^2}{4} \mathbf{I}_d + \mathbb{E}[y^{(l)} y^{(l)H}] + \frac{\Delta^2}{4} \mathbf{I}_d \\ & \quad + i \left(\mathbb{E}[x^{(l)} y^{(l)H}] + \mathbb{E}[y^{(l)} x^{(l)H}] \right) \\ &= \mathbb{E}[z^{(l)} z^{(l)H}] + \frac{\Delta^2}{2} \mathbf{I}_d, \end{aligned} \quad (6)$$

and thus

$$\mathbb{E}[\dot{z}_\Omega^{(l)} \dot{z}_\Omega^{(l)H}] = \mathbb{E}[z_\Omega^{(l)} z_\Omega^{(l)H}] + \frac{\Delta^2}{2} \mathbf{I}_{|\Omega|}. \quad (7)$$

This implies that the *quantized sample covariance matrix (Q-SCM)*

$$\hat{R}_\Omega = \sum_{l=1}^n \dot{z}_\Omega^{(l)} \dot{z}_\Omega^{(l)H} - \frac{\Delta^2}{2} \mathbf{I}_{|\Omega|} \quad (8)$$

is an unbiased estimator of the sub-covariance matrix $T_\Omega = \mathbb{E}[z_\Omega z_\Omega^H]$.

To recover the full covariance matrix from $\hat{R}_\Omega$, we exploit the Toeplitz structure of T and apply a Toeplitz projection. Note that (6) implies that, in expectation, the difference between the second-order moment of the quantized data $(\dot{z}_j, \dot{z}_k)$ and that of the unquantized data (z_j, z_k) is exactly the additive term $\frac{\Delta^2}{2} \delta_{j,k}$ induced by quantization. We thus define the per-sample estimator

$$\hat{\gamma}_s^{(l)} = \frac{1}{|\Omega_s|} \sum_{(j,k) \in \Omega_s} \dot{z}_j^{(l)} \dot{z}_k^{(l)*} - \frac{\Delta^2}{2} \delta_s, \quad s \in [d] \quad (9)$$

where $\Omega_s = \{(j, k) \in \Omega \times \Omega : k - j = s\}$, and the overall estimator

$$\hat{\gamma}_s = \frac{1}{n} \sum_{l=1}^n \hat{\gamma}_s^{(l)}, \quad s \in [d]. \quad (10)$$

To ensure that each $\hat{\gamma}_s$ is well defined, we require that Ω_s is nonempty for every $s \in [d]$. Thus, we introduce the concept of a *ruler* as follows [1].

Definition 1: A subset Ω is called a *ruler* if for any $s \in [d]$, there exist indices $j, k \in \Omega$ such that $k - j = s$.

Under a ruler Ω , the corresponding Toeplitz covariance estimator is then defined as

$$\hat{\mathbf{T}} = \mathcal{T}(\hat{\gamma}_0, \hat{\gamma}_1, \dots, \hat{\gamma}_{d-1}). \quad (11)$$

We refer to $\hat{\mathbf{T}}$ as the *quantized Toeplitz-projected sample covariance matrix (Q-TSCM)*. It can be verified that $\hat{\mathbf{T}}$ is an unbiased estimator of $\mathbf{T}$, as stated below.

Theorem 2: Fix a quantization level $\Delta \geq 0$ and a ruler Ω . The estimators $\hat{\gamma}_s$ defined in (10) and $\hat{\mathbf{T}}$ defined in (11) are unbiased, i.e.,

$$\mathbb{E}[\hat{\gamma}_s] = \gamma_s, \quad s \in [d], \quad \text{and} \quad \mathbb{E}[\hat{\mathbf{T}}] = \mathbf{T}. \quad (12)$$

Proof. It follows directly from (7), together with the definitions of Ω_s , $\hat{\gamma}_s^{(l)}$. ■

Notably, Theorem 2 imposes no specific distributional assumption on $\mathcal{D}$. It follows from Theorem 2 that Q-TSCM provides a computationally efficient and effective estimator of the true covariance matrix $\mathbf{T}$.

B. Q-SPA

The Q-TSCM exploits the Toeplitz structure; however, it is not guaranteed to be positive semidefinite in general, which might be of crucial concern in applications. In the case of unquantized data, existing algorithms, e.g., SPA [12], have been proposed to utilize the Toeplitz and positive semidefiniteness structures simultaneously to improve the estimation accuracy. Motivated by SPA, we propose an analogous algorithm, termed the *quantized SPA (Q-SPA)*, for Toeplitz covariance estimation from quantized incomplete data.

Note that $\hat{\mathbf{R}}_\Omega$ provides an unbiased estimator of $\mathbf{T}_\Omega + \frac{\Delta^2}{2} \mathbf{I}_{|\Omega|}$. Adopting the covariance-fitting criterion used in SPA and accounting for quantization effects, we consider the following formulation

$$\begin{aligned} \min_{\mathbf{u}} \quad & \left\| \hat{\mathbf{R}}_\Omega^{-\frac{1}{2}} (\hat{\mathbf{R}}_\Omega - \dot{\mathbf{R}}_\Omega) \hat{\mathbf{R}}_\Omega^{-\frac{1}{2}} \right\|_F^2, \\ \text{s.t.} \quad & \dot{\mathbf{R}} = \mathcal{T}(\mathbf{u}) \succeq \frac{\Delta^2}{2} \mathbf{I}_d. \end{aligned} \quad (13)$$

The objective is adopted from [11] and subsequently used in [12] as a surrogate for the weighted least-squares metric. The constraint reflects the quantization-induced bias term $\frac{\Delta^2}{2} \mathbf{I}_d$ in $\hat{\mathbf{R}}$, and ensures the positive semidefiniteness of $\mathbf{T}$.

A simple calculation shows that (13) is equivalent to

$$\begin{aligned} \min_{\mathbf{u}} \quad & \text{tr}(\hat{\mathbf{R}}_\Omega^{-1} \dot{\mathbf{R}}_\Omega) + \text{tr}(\dot{\mathbf{R}}_\Omega^{-1} \hat{\mathbf{R}}_\Omega), \\ \text{s.t.} \quad & \dot{\mathbf{R}} = \mathcal{T}(\mathbf{u}) \succeq \frac{\Delta^2}{2} \mathbf{I}_d. \end{aligned} \quad (14)$$

Using the identity

$$\text{tr}(\mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X}) = \min_{\mathbf{U}} \text{tr}(\mathbf{U}) \quad \text{s.t.} \quad \begin{pmatrix} \mathbf{U} & \mathbf{X}^H \\ \mathbf{X} & \mathbf{Y} \end{pmatrix} \succeq \mathbf{0}, \quad (15)$$

the optimization problem (14) can be reformulated as a semidefinite program (SDP)

$$\begin{aligned} \min_{\mathbf{u}, \mathbf{U}} \quad & \text{tr}(\hat{\mathbf{R}}_\Omega^{-1} \dot{\mathbf{R}}_\Omega) + \text{tr}(\mathbf{U}), \\ \text{s.t.} \quad & \begin{pmatrix} \mathbf{U} & \hat{\mathbf{R}}_\Omega^{\frac{1}{2}} \\ \hat{\mathbf{R}}_\Omega^{\frac{1}{2}} & \dot{\mathbf{R}}_\Omega \end{pmatrix} \succeq \mathbf{0}, \\ & \dot{\mathbf{R}} = \mathcal{T}(\mathbf{u}) \succeq \frac{\Delta^2}{2} \mathbf{I}_d, \end{aligned} \quad (16)$$

which can be solved using standard SDP solvers (e.g., CVX). Once the solution $\check{\mathbf{u}}$ to $\mathbf{u}$ is obtained from (16), we define

$$\check{\mathbf{T}} = \mathcal{T}(\check{\mathbf{u}}) - \frac{\Delta^2}{2} \mathbf{I}_d \quad (17)$$

as an estimator of $\mathbf{T}$. We refer to $\check{\mathbf{T}}$ as the Q-SPA estimator, in view of its similarity to SPA in [12].

It is shown in [3] that SPA can also be interpreted as computing a weighted atomic norm of the square root of the covariance matrix, with the weight determined by the covariance matrix itself. Therefore, SPA promotes sparse frequency recovery, which corresponds to estimating a low-rank Toeplitz covariance matrix under the Vandermonde decomposition. Consequently, one may expect Q-SPA to yield improved covariance estimates, especially when the underlying Toeplitz covariance matrix $\mathbf{T}$ is (approximately) low-rank. Numerical results will be provided to validate its improved accuracy compared to Q-TSCM, at the cost of increased computational complexity due to solving the SDP.

IV. SIMULATION RESULTS

In our numerical experiments, we randomly generate the true covariance matrix $\mathbf{T}$. Specifically, we

- 1) sample d distinct frequencies from $\mathcal{U}(0, 1)$ and construct the corresponding Fourier matrix $\mathbf{F}_d \in \mathbb{C}^{d \times d}$;
- 2) sample d independent realizations from $\mathcal{N}(0, 1)$, take their absolute values as amplitudes, and construct the diagonal matrix $\mathbf{D}$;
- 3) construct a random Toeplitz covariance matrix $\mathbf{T}$ by using the Vandermonde decomposition of Toeplitz matrices [26].

We then draw n i.i.d. samples $\mathbf{z}^{(l)}, l = 1, 2, \dots, n$ from $\mathcal{D} = \mathcal{CN}(\mathbf{0}, \mathbf{T})$ and obtain their triangular-quantized sub-observations $\hat{\mathbf{z}}_\Omega^{(l)}$, which are used for covariance estimation. Each data point in our numerical results is obtained from 100 Monte Carlo simulations.

In *Experiment 1*, we study the effects of the sample size n . We fix $d = 16$, $\Delta = 5$, and consider two special rulers $\Omega_1 = \{1, 2, \dots, 16\}$ and $\Omega_{1/2} = \{1, 2, 3, 4, 8, 12, 16\}$; we then compute the relative error for varying n and report the results on a log-log scale in Fig. 1². The curves are approximately linear with slope -0.5 , implying that both Q-TSCM and Q-SPA are unbiased with convergence rate $\mathcal{O}(n^{-1/2})$. For large

²When n is small, we introduce a diagonal perturbation $\epsilon \mathbf{I}_{|\Omega|}$ to the sample covariance matrix to compute its inverse before performing Q-SPA; after all, we know a priori that its expectation $\dot{\mathbf{R}} \succeq \frac{\Delta^2}{2} \mathbf{I}_d$.

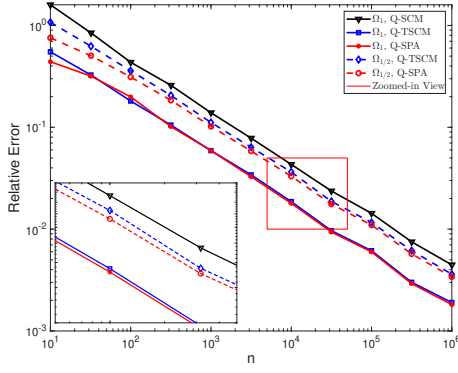


Fig. 1. Impact of sample size n on Toeplitz covariance estimation.

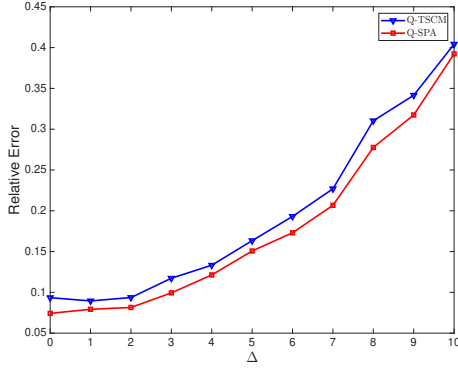


Fig. 2. Impact of quantization level Δ on Toeplitz covariance estimation.

n , Q-SPA typically achieves the best accuracy. Notably, Q-SCM performs worse than Q-TSCM and can be inferior even to the sparse-observation case based on $\Omega_{1/2}$, highlighting the substantial gains achievable by exploiting the Toeplitz structure.

In *Experiment 2*, we investigate the impact of the quantization level Δ . We set $d = 16$ and $n = 500$, and choose the sparse ruler $\Omega_{1/2}$. We then plot the relative error versus Δ for different methods in Fig. 2. It can be observed that, for both methods, the relative error increases with Δ ; it grows slowly for small Δ and then increases more rapidly as Δ becomes larger. This suggests that the impact of quantization on estimation accuracy is relatively limited within a certain range of Δ (e.g., $\Delta \leq 2$ in our setting).

In *Experiment 3*, we focus on the impact of different rulers. We set $d = 16$ and $\Delta = 5$, and report the results on a log-log scale in Fig. 3. Besides Ω_1 and $\Omega_{1/2}$, we additionally choose $\Omega_A = \{1, 2, 3, 4, 5, 6, 7, 8, 16\}$ and $\Omega_B = \{1, 2, 3, 5, 8, 11, 14, 15, 16\}$ for comparison. The estimation errors under these different rulers all decrease as n increases. Among the four choices, Ω_1 achieves the smallest errors, whereas $\Omega_{1/2}$ yields the largest errors, as $\Omega_{1/2}$ has the smallest cardinality. Moreover, although Ω_A and Ω_B have the same cardinality, estimation based on Ω_B consistently yields smaller errors. This indicates that different rulers can lead to markedly different estimation errors, depending not only on the cardinality but also on the spatial distribution of the

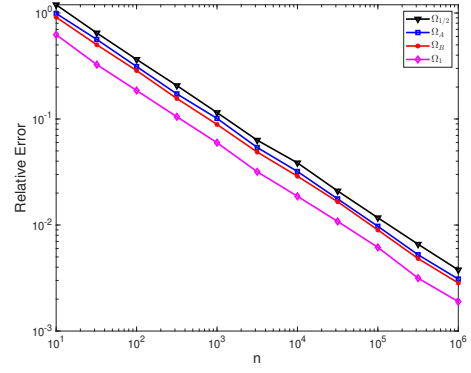


Fig. 3. Impact of different ruler Ω on Toeplitz covariance estimation.

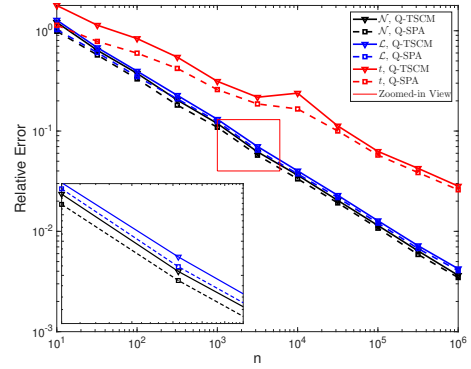


Fig. 4. Performance of Covariance Estimation under Various Distributions.

observed indices.

In *Experiment 4*, we additionally investigate the performance of covariance estimation under different distributions $\mathcal{D}$. We set $d = 16$, $\Delta = 5$, and choose the sparse ruler $\Omega_{1/2}$. The relationship between the relative error of covariance estimation and the sample size n under different distributions is plotted in Fig. 4. We consider the normal distribution ($\mathcal{N}$), the Laplace distribution ($\mathcal{L}$), and the t -distribution with $\nu = 3$ degrees of freedom for comparison. It can be seen that the Laplace distribution yields performance close to that under the normal distribution; however, under the extremely heavy-tailed t -distribution, the relative error is noticeably larger for a given n . Nevertheless, the relative error decreases with n for all distributions, while the decay is slower in the heavy-tailed setting.

V. CONCLUSION

Under sparse observation and coarse quantization, we propose two estimators within the triangular quantization framework, termed Q-TSCM and Q-SPA. In particular, Q-TSCM is unbiased and computationally efficient, while Q-SPA additionally enforces positive semidefiniteness and exhibits improved empirical performance. Numerical experiments demonstrate that both estimators enable accurate covariance estimation.

For a detailed performance analysis, including the bit-budget setting, we refer readers to the extended journal version [27].

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